Mini Project 2: The Wall Crossing Challenge

1. Objective

In this mini project, a swarm of e-puck robots must move from one side of an arena to the other side by passing through a wall with unknown passages. The wall contains a fixed but unknown number of holes, and each hole is narrow enough that only one robot can pass through it at a time.

The goal of this mini project is to design and implement a swarm-based robotic solution in which the robots collectively pass through the wall as efficiently as possible using local sensing, decentralized decision making, and swarm-intelligent behavior.

2. Story and Robotic Setting

A swarm of e-puck robots is placed in a closed $2\,\text{m} \times 3\,\text{m}$ arena. The robots are spawned in random positions in the left half of the arena. The arena is divided into two parts by a wall made of obstacles. Each side of the arena has an approximate size of $2\,\text{m} \times 1.5\,\text{m}$.

The wall contains k holes. The value of k is fixed during each simulation but unknown to the robots. Each hole has a width of approximately 1.5 times the robot diameter. Therefore, the holes act as physical bottlenecks, and only one robot can pass through each hole at a time.

A light source is placed on the right side of the arena. Robots may use their light sensors as a navigation signal and as a local cue for detecting that they have reached the target side.

The mission is completed when the last robot has passed through the obstacle wall and reached the right side of the arena. The performance objective is to minimize the total mission completion time.

3. Implementation Platform

The default implementation platform for this mini project is **Webots**, using **e-puck robots**.

Subject to approval by the lecturer, students may use another robotics simulation platform. In that case, the alternative platform must still satisfy the same project requirements, including local robot sensing, no external localization in the robot controller, wall passage constraints, and swarm-based coordination.

4. Implementation Requirements

4.1. Mandatory Technical Requirements

The implementation shall satisfy all of the following:

- The robot controller shall not use external localization, global position information, or the Webots supervisor view.
- The robot controller shall rely only on the robot's own sensing and local interaction capabilities.
- If robot-to-robot communication is implemented, it shall be limited to nearby robots within a communication range of 1 m.

- At least one explicit swarm-intelligence mechanism shall be implemented.

4.2. Recommended Default Environment

The following default setting are recommended:

- Number of robots: $6 \leq n \leq 10$
- Number of holes: $2 \leq k \leq 4$

5. Deliverables and Grading

5.1. Deliverables

Each group shall submit:

- a fully functional implementation of the robotic simulation,
- a short report (max two pages) describing:
  - the implemented approach,
  - the swarm-intelligence mechanism used,
  - the simulation setup,
  - the obtained results,
  - the main insights and lessons learned.

5.2. Group Setup

The project must be completed in groups of 2 to 3 students. Solo project development is possible only when necessary and must be approved by the lecturer.

5.3. Grading

The normal project grade consists of:

- **Running implementation** (50%)
- **Quality of the swarm-based algorithm with respect to swarm robotics principles** (25%) [report]
- **Evaluation and results** (25%) [report]

5.4. Optional Leaderboard Bonus

Volunteer groups may participate in a leaderboard competition after the project submission deadline. The leaderboard setup and scoring criteria will be announced separately.

The leaderboard can provide up to 20% bonus points. Therefore, the maximum achievable score is:

$$100\% \text{ normal grade} + 20\% \text{ leaderboard bonus} = 120\%.$$